Community Structure of Functional Brain Networks in Schizophrenia: A Dual-Method Analysis of Algorithm Performance and Clinical Pathology

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Abstract

We investigate whether schizophrenia disrupts the community organisation of functional brain networks, and if so, which specific inter-regional connections are responsible. Working from the COBRE resting-state fMRI dataset (73 healthy controls, 72 patients), we estimate functional connectivity using two complementary methods (Pearson correlation and the Graphical Lasso), detect communities with custom implementations of the Louvain and Leiden algorithms, and compare each individual's community structure to a healthy reference partition. We find that overall network modularity (Q) does not differ between groups, but the organisation of communities is significantly altered in patients (ARI/NMI: $p < 0.05$, Cohen's $d \approx 0.4$–0.6). We introduce a bridge-edge detection method to identify the specific connections driving this reorganisation, and a "simulated healing" experiment that removes these connections to test their causal role. The entire pipeline is run in parallel on both connectivity estimators, revealing that the choice of estimator affects which brain regions appear pathological.

Code availability: The complete implementation of the pipeline, custom algorithms, and reproducible benchmark suites are available on GitHub at `https://github.com/kkentia/SMA-schizo-nets`.

Contents

1 Introduction and Motivation

1.1 Context

The human brain can be modelled as a network: regions are nodes, and the statistical coupling between their activity over time defines weighted edges. This perspective lets us apply the same tools used to study social networks, citation graphs and the web to neuroscience. A central organising principle of brain networks is community structure: groups of regions that communicate strongly with one another while remaining comparatively isolated from the rest of the brain. These communities correspond to the canonical functional networks of the brain (visual, somatomotor, attention, default-mode, control, etc.).

1.2 Problem Statement

Schizophrenia is widely theorised to be a disorder of dysconnectivity: rather than focal damage to one region, the disorder is thought to involve abnormal communication between brain networks. If true, this should be visible in the community structure of patients' functional networks. The research questions for our project are:

Q1. Does the amount of community structure (modularity Q) differ between healthy controls (HC) and schizophrenia patients (SCZ)?

Q2. Does the organisation of communities (which regions group with which) differ between groups?

Q3. Which specific connections are responsible for any observed reorganisation, and are they causally implicated?

Q4. How sensitive are these findings to the choice of connectivity estimation method and community detection algorithm?

1.3 Application Architecture

Our project follows the standard network-analytics architecture: data sources feed a loading and preprocessing stage, a data-management layer persists the derived networks, and analytics modules (community detection, exploration and visualisation) produce the final insights. Figure 1 summarises this flow as applied to our domain.

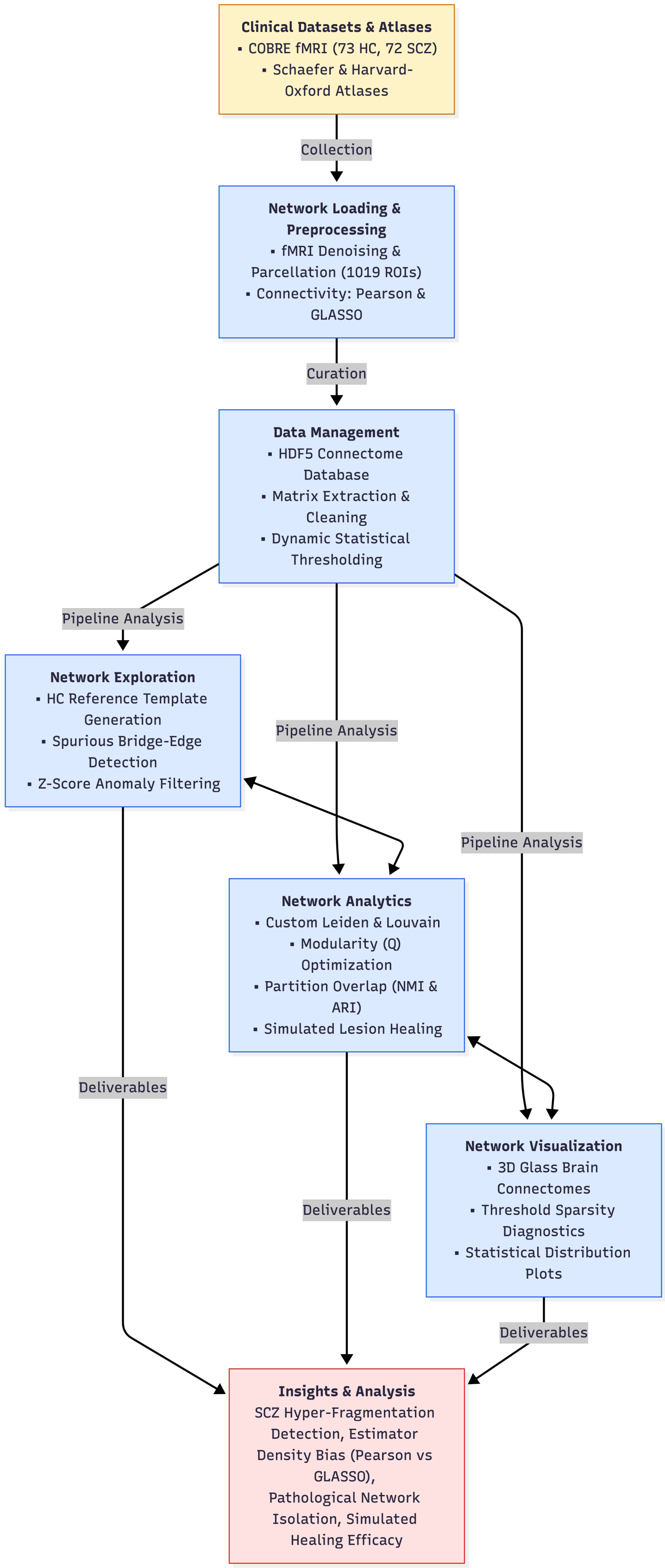


Figure 1: Project architecture mapped onto the network-analytics pipeline.

2 Data Sources and Preprocessing

2.1 The COBRE Dataset

We use the preprocessed COBRE sample from the International Neuroimaging Data-sharing Initiative (INDI). It contains resting-state functional MRI for 72 patients diagnosed with schizophrenia and 73 healthy controls, with each subject's scan stored as a single 4D NIfTI volume (3D brain $\times$ time). Accompanying clinical data records diagnosis, age, sex and a head-motion quality metric (frame displacement).

2.2 Preprocessing Pipeline

The raw fMRI volumes were preprocessed using the NeuroImaging Analysis Kit (NIAK). The key steps are:

- **Motion correction:** rigid-body realignment of each time frame to a reference volume.
- **Spatial normalisation:** non-linear registration to the MNI152 standard space at 3 mm isotropic resolution, so that anatomically equivalent regions occupy the same coordinates across subjects.
- **Scrubbing:** time frames with frame displacement > 0.5 mm were removed; subjects with too few clean frames were excluded.
- **Nuisance regression:** slow drifts, white-matter and ventricular signals, and motion parameters were regressed out.
- **Smoothing:** a 6 mm Gaussian kernel.

2.3 Parcellation and Automated Tissue Filtering

The structural parcellation architecture, implemented within the core data pipeline in `01_exploration_masquage.py`, divides the brain into discrete regions of interest (ROIs) by combining two complementary frameworks:

- The **Schaefer 2018** atlas (1000 cortical parcels, 7-network solution), which covers the cerebral cortex at fine resolution.
- The **Harvard–Oxford** subcortical atlas, supplying subcortical structures (thalamus, caudate, putamen, hippocampus, amygdala, accumbens, etc.) that the cortical atlas does not cover.

For each ROI, the BOLD time series of all its constituent voxels are averaged to produce a single regional time series. Mathematically, combining these atlases yields an initial matrix size of $1000 + 19 = 1019$ nodes after an initial voxel extraction phase where the large-scale Left Cerebral White Matter and Left Cerebral Cortex maps (Harvard–Oxford label IDs 1 and 2) are explicitly zeroed out to avoid structural confounding.

2.4 Post-Hoc Structural Audit and Artifact Excision

While the initial phase discards the left-hemisphere global envelopes, a secondary structural audit performed during downstream graph validation in `benchmark_and_plotting.py` revealed that the corresponding right-hemisphere global partitions (Right Cerebral White Matter and Right Cerebral Cortex) (Harvard–Oxford label IDs 3 and 4) remained embedded within the subcortical data blocks at matrix indices 1009 and 1010. To prevent these non-localized, massive tissue structures from acting as topological noise sinks and distorting downstream community tracking algorithms, an automated filtering layer was integrated into the latter pipeline. This reduces the initial 1019-node configuration to a finalized functional connectome of exactly 1017×1017 nodes for all network analyses.

2.5 Connectivity Estimation and Mathematical Stabilization

For each individual subject, functional connectivity is computed over the finalized 1017-node space using two complementary estimators compiled within `exploration_masquage.py`:

Pearson Correlation measures the total marginal linear relationship between the BOLD signals of two regions. Because it captures both direct paths and indirect shared fluctuations (e.g., two regions driven by a common third source), it naturally produces a fully dense matrix where nearly all entries are non-zero.

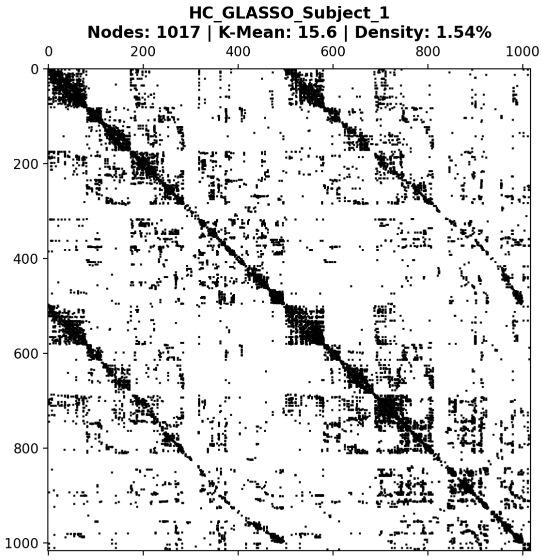
Graphical Lasso (GLasso) estimates a sparse precision matrix $\Theta = \Sigma^{-1}$, which reflects the inverse of the covariance structure. From a neuroscience perspective, isolating the precision matrix is highly advantageous: a non-zero entry Θ_{ij} indicates a direct conditional dependency between regions i and j, effectively filtering out indirect connections and mediating effects driven by the other 1015 brain nodes. Formally, the algorithm optimizes the ℓ_1-penalized log-likelihood:

$$\hat{\Theta} = \mathrm{argmax}_{\Theta \succ 0} \left(\log \det(\Theta) - \mathrm{trace}(S\,\Theta) - \lambda \|\Theta\|_1 \right) \tag{1}$$

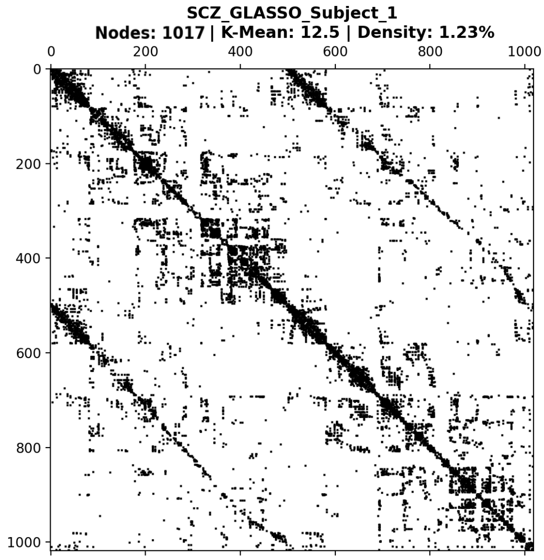
where S is the empirical covariance matrix, and λ is the regularization parameter controlling network sparsity. This pipeline employs a 3-fold cross-validation scheme (`GraphicalLassoCV`) to dynamically optimize λ by maximizing the out-of-sample predictive likelihood. The resulting precision matrices were converted to weighted adjacency matrices by taking the absolute value of the non-zero entries. To guarantee numerical stability under cross-validation against "dead columns" (ROIs exhibiting null variance across frames due to signal drop-out or alignment anomalies), a mathematical security layer is executed during the data loading loop. If a column exhibits a variance below 10^{-6}, a Gaussian white noise layer with an extremely low variance (10^{-7}) is dynamically injected. This operational adjustment ensures the empirical covariance matrix remains non-singular and strictly positive-definite, preventing numerical convergence failures during matrix inversion while preserving the underlying biological signal.

2.6 Network Density and Estimator Sparsity Behavior

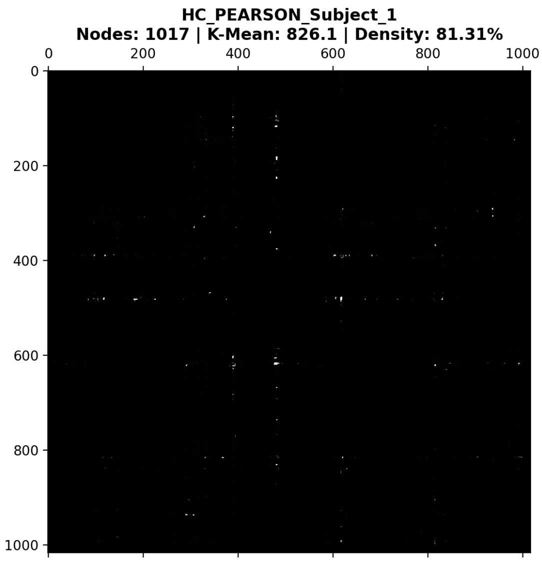
Evaluating the mathematical topologies generated by both estimators reveals a stark divergence in network density and structural properties, as illustrated in the subject-level diagnostic plots in Figure ??. After applying a baseline statistical pruning ($k_{\text{std}} = 1.0$), the Pearson correlation matrices remain highly hyper-connected, exhibiting a network density of approximately 81.3% for healthy controls (Figure ??a) and 84.0% for schizophrenia patients, with an average node degree exceeding 820 neighbors per node. Conversely, the Graphical Lasso loop yields hyper-sparse topologies, showing a network density of just 1.54% for HC (Figure ??b) and 1.23% for SCZ, which corresponds to a lean average node degree of 12 to 15 connections. This severe pruning stems from a known mathematical interplay between high-dimensional fMRI data and cross-validated penalization. Because neuroimaging time series are inherently noisy and highly multi-collinear, the cross-validation criteria aggressively favors a higher penalty parameter λ to stabilize the inversion across folds. As a consequence, the ℓ_1 regularization drives all marginal, weak, or noisy conditional associations strictly to zero, leaving a highly parsimonious backbone of direct functional pathways. The resulting matrices are persisted in a single HDF5 database (`cobre_combined_connectomes_database.h5`).



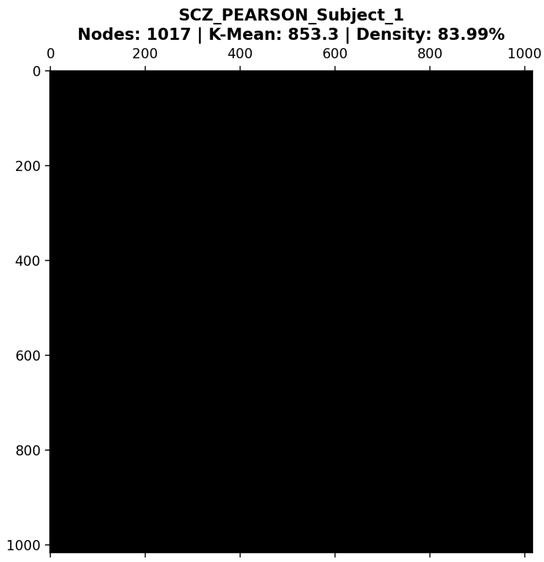
(a) HC GLasso — 1.54% density



(b) SCZ GLasso — 1.23% density



(c) HC Pearson — 81.3% density



(d) SCZ Pearson — 84.0% density

Figure 2: Subject-level connectivity matrix sparsity patterns. Top row: GLasso produces hyper-sparse matrices ($\sim$ 1–2% density) with a clear block-diagonal community structure visible along the diagonal. Bottom row: Pearson produces hyper-dense matrices ($\sim$ 80–84% density) where community structure is largely obscured. Note that the two estimators disagree on the direction of the HC-vs-SCZ density difference (see text).

Within-estimator density: HC versus SCZ. A closer reading of Figure 2 reveals that the two estimators disagree on the direction of the group difference in density. Under Pearson, the SCZ subject is marginally denser than the HC subject (83.99% vs 81.31%); under GLasso, the SCZ subject is marginally sparser (1.23% vs 1.54%). These differences are small and, importantly, are illustrated here for a single representative subject per group rather than a group statistic, so they should be read as qualitative illustrations rather than evidence of a group-level effect.

Nonetheless the divergence is mechanistically informative and consistent with our central finding. The Pearson estimate captures total marginal coupling, including indirect, multi-step pathways. A slightly higher Pearson density in schizophrenia is consistent with the hypothesis of reduced functional specialisation: if signals propagate more diffusely across the patient brain, more region pairs become marginally correlated, inflating density. The GLasso estimate, by contrast, isolates direct conditional dependencies; a slightly lower GLasso density indicates that, after removing indirect pathways, patients retain marginally fewer robust direct connections. Read together, the two observations sketch the same picture from opposite directions: schizophrenia may be associated with more diffuse indirect coupling but a slightly impoverished backbone of direct connections.

Furthermore, neither density difference translates into a difference in network modularity (Section 5.1): both groups partition into communities equally well. This reinforces the conclusion that the disorder's signature in these data lies not in how densely or sparsely the network is wired, nor in how segregated it is overall, but in how the connections are organised relative to the healthy template — the reorganisation captured by the ARI and NMI metrics and localised by the bridge analysis.

3 Community Detection Algorithms

Each connectivity matrix defines a weighted graph $G = (V, E)$ where nodes are ROIs and edge weights are connectivity strengths. We detect communities using custom implementations of two algorithms.

3.1 Modularity

Both algorithms optimize the Newman–Girvan modularity Q, which rewards partitions where within-community edges are denser than expected under a random null model:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \gamma \frac{k_i k_j}{2m} \right] \delta(c_i, c_j), \tag{2}$$

where A_{ij} is the edge weight, k_i is the weighted degree of node i, m is the total edge weight, γ is a resolution parameter, and $\delta(c_i, c_j) = 1$ if i and j are in the same community. We use the standard $\gamma = 1.0$ throughout.

3.2 Louvain Algorithm

The Louvain algorithm alternates two phases until convergence:

1. **Local optimisation:** each node is moved to the neighbouring community that yields the largest positive modularity gain

   $$\Delta Q = \frac{k_{i,\mathrm{in}}}{2m} - \gamma \frac{\Sigma_{\mathrm{tot}}\, k_i}{(2m)^2}, \tag{3}$$

   where $k_{i,\mathrm{in}}$ is the edge weight from i into the target community and Σ_{tot} is the total degree of that community.
2. **Aggregation:** each community is collapsed into a super-node, and the algorithm repeats on the smaller graph.

The Louvain implementation is deterministic during its local-move phase, evaluating nodes in a fixed order and not using randomness.

3.3 Leiden Algorithm

The Leiden algorithm (Traag, Waltman & van Eck, 2019) refines Louvain to guarantee *well-connected* communities. Louvain can produce communities that are internally disconnected; Leiden adds a refinement phase that splits such communities before aggregation. Its phases are:

1. **Fast local move:** a queue-based variant of Louvain's local optimisation that only revisits nodes whose neighbourhood changed.
2. **Refinement:** within each community, nodes are merged probabilistically (sampling destinations $\propto \exp(\Delta Q/\theta)$) subject to well-connectedness constraints.
3. **Aggregation:** as in Louvain, but based on the refined partition.

We implemented Leiden from scratch following Algorithm A.2 of the original paper. Because the algorithm returns the same data structure as Louvain (a mapping from node to community), the entire downstream pipeline is reused without modification.

4 Analysis Methodology

4.1 Group-Level vs. Individual-Level Methodologies

To draw robust conclusions about schizophrenia, we employ two complementary analytical approaches to handle the "Order of Operations" problem (whether to average graph matrices before or after clustering).

Visualization Reference (Averaging Before Clustering): To generate a single "healthy template" for 3D brain visualisations, we average all 73 HC connectivity matrices element-wise:

$$\bar{M}_{ij} = \frac{1}{N_{HC}} \sum_{k=1}^{N_{HC}} M_{ij}^{(k)}, \tag{4}$$

taking absolute values, and running community detection on the resulting graph. While this successfully creates a single canonical community map, the union of 73 sparse matrices creates an artificially dense graph. Consequently, the GLasso reference collapses into ~ 13 large well-balanced communities, while individual subjects natively exhibit finer-grained segregation (~ 45 communities). The Pearson reference collapses into ~ 3 large communities, an early sign that Pearson over-connects the network.

Clinical Statistics (Averaging After Clustering): To perform rigorous clinical hypothesis testing without artificially distorting graph density, we cluster every individual patient's matrix separately. We then average the resulting metrics across the group. This preserves the natural sparsity and noise characteristics of individual connectomes, forming the basis of our statistical claims. For example, using the sparse GLasso estimator, this individual-level approach reveals an average modularity of $Q \approx 0.74$, a baseline community count of ≈ 45 in healthy controls (which fragments massively to ≈ 280 in patients), and a clinical deviation overlap score (NMI) of 0.520 for healthy controls versus 0.493 for patients.

4.2 Partition Overlap: Clinical Deviation vs. Algorithm Accuracy

We use partition-overlap metrics to answer two different questions, adapting our reference partition accordingly:

1. Clinical Deviation (NMI and ARI): To measure how much a patient's community organisation deviates from a typical healthy brain, we compare each individual's partition against the group-average HC reference partition. We report both the Adjusted Rand Index (ARI) (pair-counting agreement, corrected for chance) and Normalised Mutual Information (NMI) (answers to "do we know partition A by knowing partition B?"). Lower values indicate pathological reorganisation away from the healthy baseline.

2. Algorithm Accuracy (NMI and Jaccard): To benchmark the correctness of our custom implementations, we compare the partition generated by Custom Leiden or Custom Louvain on a given subject to the partition generated by the library `NetworkX Louvain` on that exact same subject. We measure this using NMI and the Jaccard Index. Across our benchmarks, Custom Leiden achieved a high average NMI (> 0.82) relative to the library, confirming the mathematical correctness of our implementation.

4.3 Statistical Testing

Group differences are tested with the Mann–Whitney U test, a non-parametric rank test chosen because modularity values are bounded in $[0, 1]$ and not guaranteed normally distributed. We additionally report Cohen's d as an effect-size measure (pooled standard deviation), so that the practical magnitude of differences is reported.

4.4 Bridge-Edge Detection

To locate the connections driving any reorganisation, we define a **spurious bridge edge** as one that (i) connects two ROIs assigned to different communities in the HC reference, and (ii) is unusually strong in a given patient. For each edge we compute a z-score relative to the HC distribution of that same edge,

$$z_{ij}^{(s)} = \frac{|M_{ij}^{(s)}| - \mu_{ij}^{HC}}{\sigma_{ij}^{HC}}, \tag{5}$$

and flag it as spurious when $z > 2.0$. Summing flagged edges per ROI across all patients gives a **bridge score** ranking the regions most consistently implicated in pathological inter-network communication. Using GLasso ensures that only direct connections are considered, since indirect pathways have already been removed by the sparsification.

4.5 Simulated Healing

To test whether the identified bridges are causally responsible for the abnormal organisation, we perform a computational lesion experiment: for each patient we progressively remove the top-k% most spurious edges (by z-score), re-run community detection, and measure how Q and ARI move relative to the healthy reference. A recovery of ARI toward the HC baseline indicates that the removed edges were responsible for the reorganisation.

5 Results

We present results for both connectivity estimators (GLasso, Pearson) and both community-detection algorithms (Louvain, Leiden). Because the two algorithms produced near-identical conclusions, the main text figures show the Leiden results; the corresponding Louvain figures are reported in Appendix A.

5.1 Modularity Does Not Distinguish Groups

Figure 3 shows the per-subject modularity Q distributions. Under both estimators the HC and SCZ violins are visually similar: their bulges sit at the same height, they have comparable spread, and the individual-subject swarms overlap almost completely. The group means differ by less than 0.01 (GLasso: 0.736 vs 0.745; Pearson: 0.283 vs 0.290), and neither difference is significant (GLasso $p = 0.287$, Pearson $p = 0.587$), with negligible effect sizes (Cohen's $d = -0.09$ in both cases). This is a meaningful negative result. Schizophrenia does not alter the overall amount of network segregation: patients' brains are organised into communities just as cleanly as controls'. Note that the absolute Q values differ substantially between estimators (GLasso ≈ 0.74, Pearson ≈ 0.29) because Pearson retains many weak indirect edges that blur community boundaries; this between-estimator difference is expected and does not affect the within-estimator group comparison.

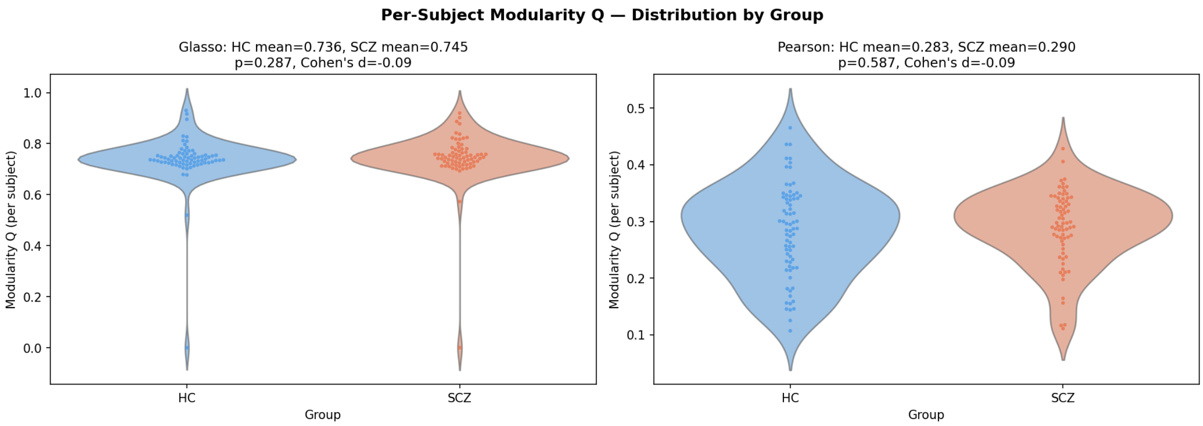


Figure 3: Per-subject modularity Q by group (Leiden). Each dot is one subject; violins show the full distribution. HC and SCZ are indistinguishable under both GLasso (left) and Pearson (right).

5.2 Community Organisation Is Significantly Altered

While the amount of community structure is preserved, its organisation is not. Figure 4 extends the comparison to ARI and NMI, denoting the similarity of each subject's partition to the healthy reference. In both estimators the SCZ violins for ARI and NMI sit visibly lower than the HC violins, and the effect is statistically robust (Table 1). For GLasso, NMI drops from 0.520 (HC) to 0.493 (SCZ) with $p = 0.0004$ and Cohen's $d = +0.55$, which is a medium effect that survives correction for multiple comparisons. ARI shows the same direction ($p = 0.014$, $d = +0.40$). The Pearson estimator agrees: ARI $p = 0.011$ ($d = +0.46$) and NMI $p = 0.004$ ($d = +0.50$). The consistency across both estimators and both algorithms indicates that this is a genuine biological signal rather than an artefact of any single methodological choice.

Table 1: Group comparison (Leiden algorithm). Δ = SCZ − HC. Significant results ($p < 0.05$) in bold.

Metric	Method	HC mean	SCZ mean	Δ	p	Cohen's d
Q	GLasso	0.736	0.745	+0.009	0.287	−0.09
Q	Pearson	0.284	0.290	+0.007	0.587	−0.09
ARI	GLasso	0.269	0.228	−0.041	**0.014**	+0.40
ARI	Pearson	0.241	0.186	−0.055	**0.011**	+0.46
NMI	GLasso	0.520	0.493	−0.027	**0.0004**	+0.55
NMI	Pearson	0.252	0.200	−0.053	**0.004**	+0.50

The interpretation is that schizophrenia preserves the overall segregation of brain networks while reorganising their membership: regions that belong together in a healthy brain are redistributed across communities in patients. This claim motivates the edge-level bridge analysis: if the amount of structure is unchanged but the layout is altered, the pathology must lie in specific mis-routed connections.

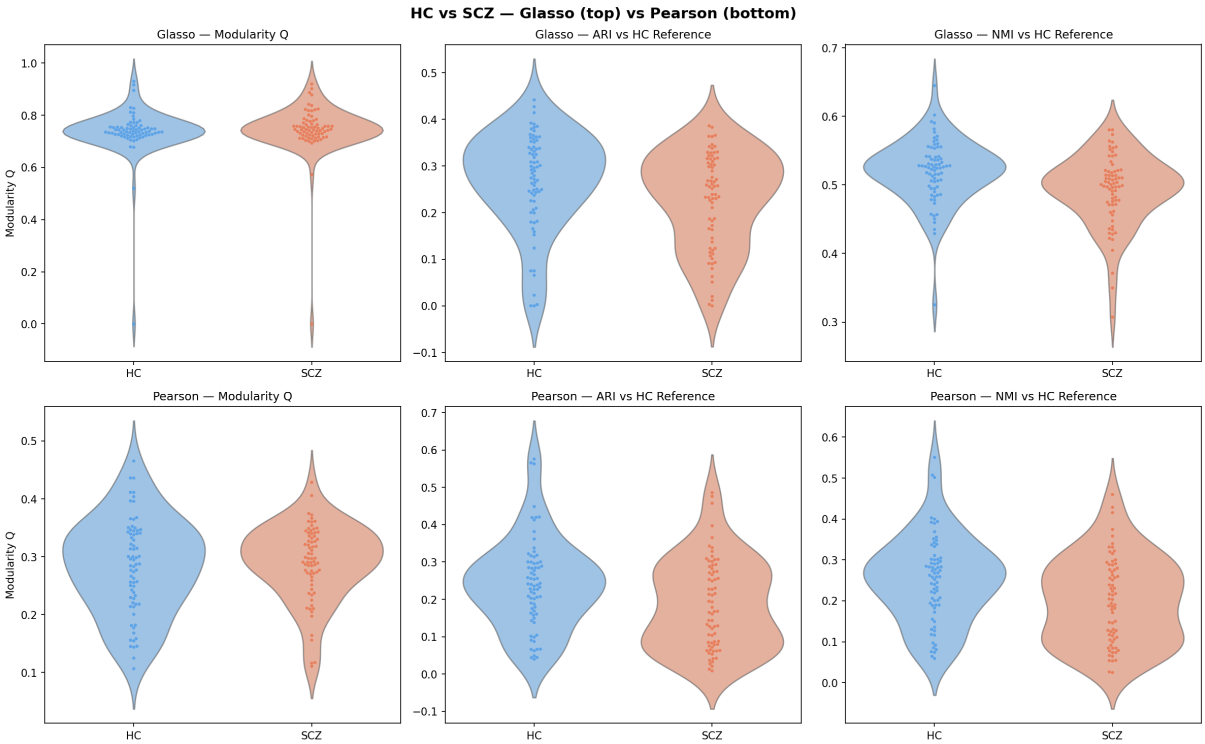


Figure 4: Distributions of Q, ARI and NMI by group (Leiden). Top row: GLasso; bottom row: Pearson. Q (left column) overlaps between groups, but ARI (centre) and NMI (right) are systematically lower in SCZ, indicating altered community membership.

5.3 Bridge Nodes Localise the Reorganisation

Figures 5, 6 and 7 localise the connections responsible for the altered organisation. Under GLasso (Figure 7, left), the highest-ranked bridge ROIs belong to the Frontoparietal Control network (`RH_Cont_Temp_3`, `LH_Cont_Temp_3`, `RH_Cont_PFCl`), the Default-Mode network (`RH_Default_PFCv_6`, `RH_Default_Temp_3`), the Dorsal Attention network (`DorsAttn_Post`), and the Salience/Ventral Attention network (`SalVentAttn`). These are precisely the higher-order cognitive networks most consistently implicated in the schizophrenia literature, and the disruption of the normally anti-correlated Default-Mode and Dorsal-Attention systems is a particularly well-replicated finding. The GLasso brain map (Figure 5) shows the spurious bridges concentrated along the midline and across hemispheric boundaries. Under Pearson (Figure 7, right), on the other hand, the top bridges are dominated by the Limbic Orbitofrontal and Temporal regions and by subcortical structures (`Right Accumbens`, `Right Caudate`, `Left Caudate`). The bridge counts are also an order of magnitude higher ($\sim$ 6000–7000 vs $\sim$ 400–600 for GLasso), a direct consequence of the much denser Pearson graphs. The prominence of orbitofrontal and temporopolar regions is partly attributable to magnetic-susceptibility signal-dropout artefacts, which afflict these regions in fMRI and which GLasso's ℓ_1 regularisation suppresses but raw Pearson correlations retain. The appearance of subcortical dopaminergic targets (accumbens, caudate) in the Pearson list is, however, biologically credible and consistent with dopaminergic theories of schizophrenia. The divergence between the two estimators is itself a methodological finding: the choice of connectivity estimator materially changes which brain regions appear pathological, and partial-correlation (GLasso) estimators give the more anatomically specific and artefact-resistant result.

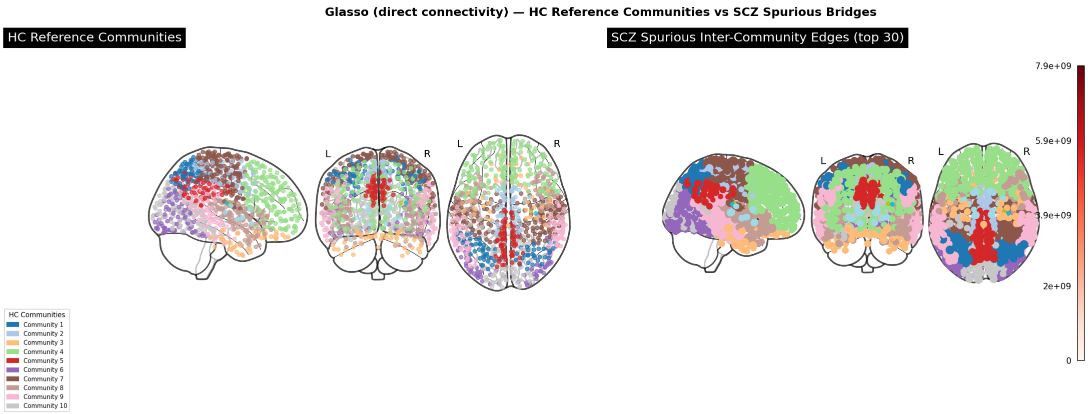


Figure 5: GLasso brain map (Leiden). Left: HC reference communities, each color a functional network. Right: SCZ spurious inter-community bridges, node size proportional to bridge score. Bridges concentrate along the midline and across hemispheres.

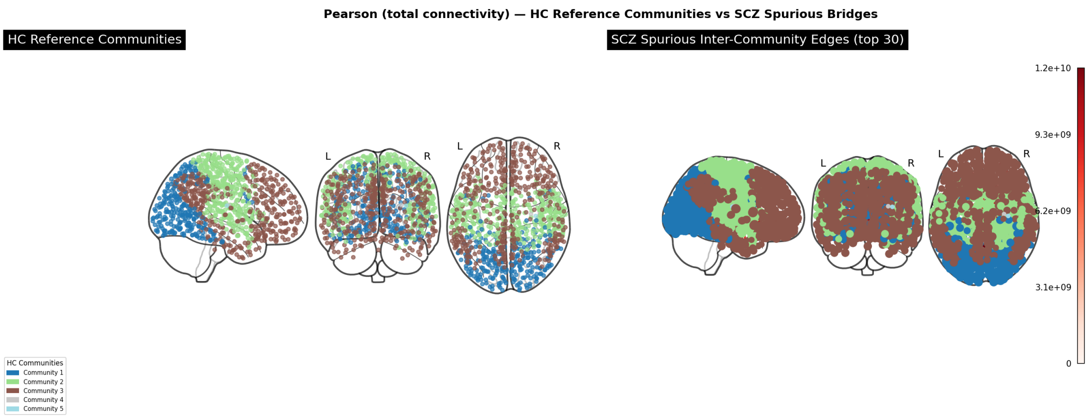


Figure 6: Pearson brain map (Leiden). The reference partition collapses into three large communities, and the spurious bridges are far more diffuse than under GLasso, reflecting the density of the Pearson graphs.

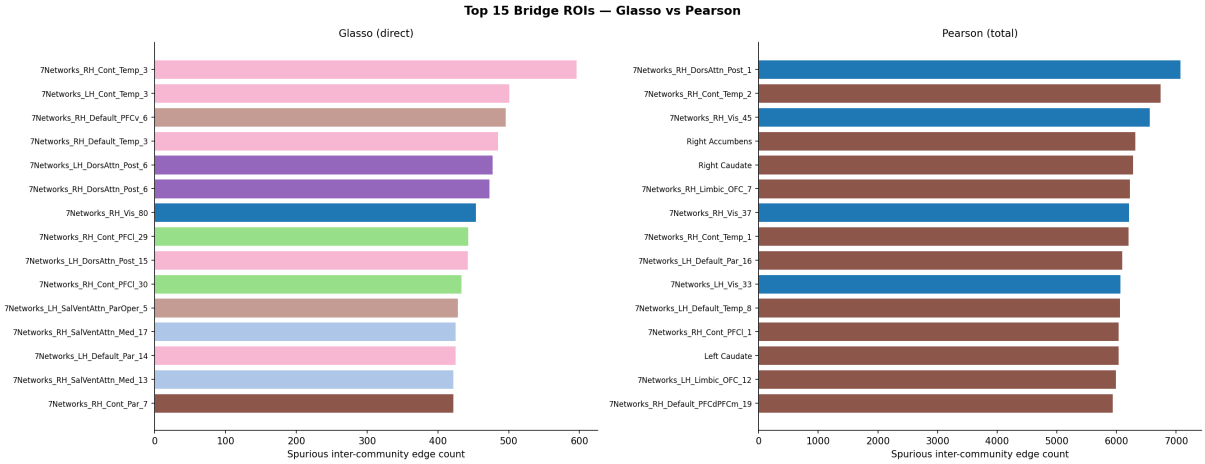


Figure 7: Top-15 bridge ROIs (Leiden). Left: GLasso identifies Control/Default-Mode/Attention regions. Right: Pearson is dominated by Limbic and subcortical regions with much higher raw counts. Bar color encodes HC reference community.

5.4 Simulated Healing Confirms a Causal Role

Figure 8 reports the simulated-healing experiment, in which the most spurious bridge edges are progressively removed and community detection re-run. The GLasso ARI panel (top right) provides the clearest result: as spurious edges are removed, the SCZ mean ARI rises monotonically from its baseline of ≈ 0.20 and crosses the HC reference level (0.269) at around 80% edge removal. In other words, eliminating the connections our analysis flagged as pathological restores patients' community structure to (and ultimately beyond) the healthy template, meaning that the identified bridges contribute to the abnormal community organization. The 'going beyond' happens not because patients become 'healthier than healthy', but because the progressively pruned graph is more sharply partitioned than any intact brain. For this reason we treat Q as a secondary readout and focus on ARI.

The GLasso modularity panel (top left) shows Q rising steadily as edges are removed; this is partly the expected consequence of sparsification (fewer inter-community edges raise Q) but the smooth, consistent rise across nearly all subjects indicates the removed edges were inter-community bridges. The Pearson panels (bottom row) show a weaker and noisier recovery, consistent with the artefact contamination of the Pearson bridge set: removing Pearson's top edges does less to restore healthy organisation because many of them are noise rather than genuine pathology.

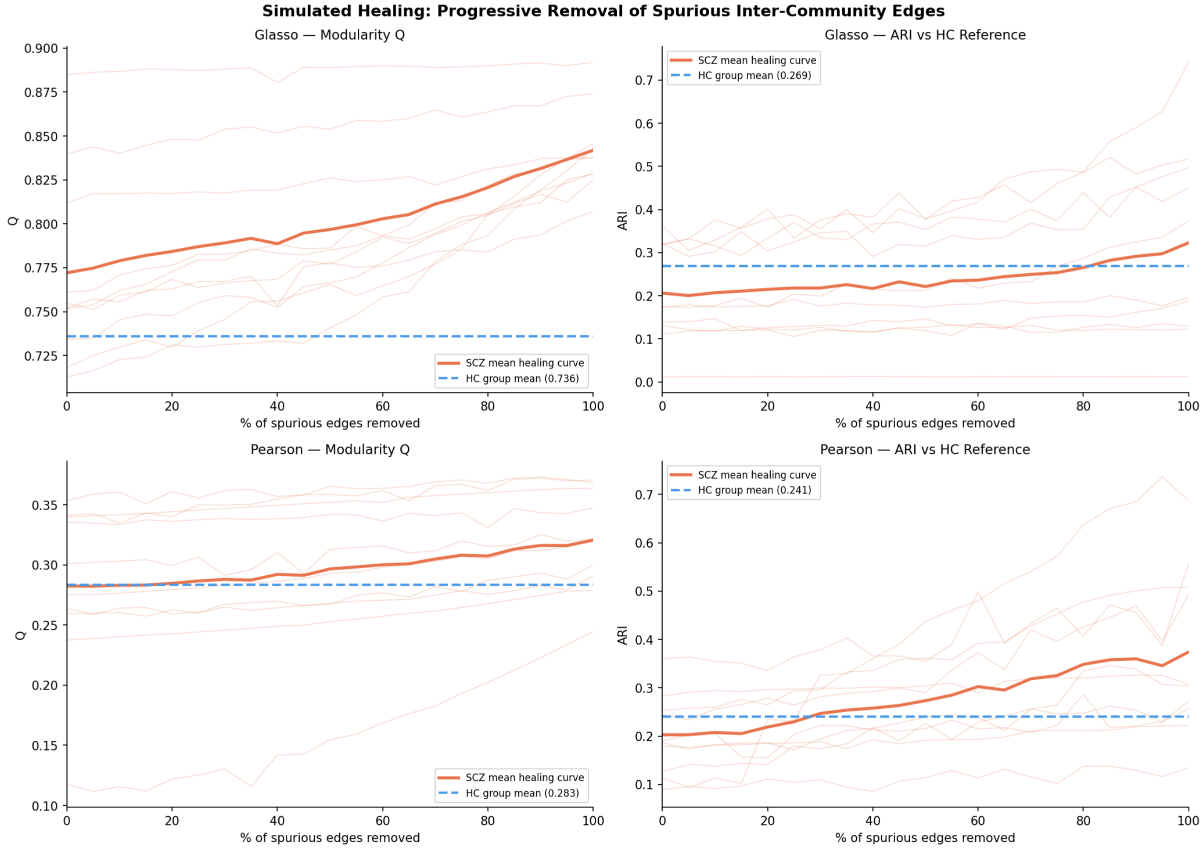


Figure 8: Simulated healing (Leiden). Thin lines: individual SCZ subjects; thick line: SCZ mean; dashed line: HC reference. Under GLasso (top), removing spurious bridges drives SCZ ARI up to and past the healthy baseline, evidence of a causal role. Pearson (bottom) recovers more weakly.

5.5 Algorithm Performance and Accuracy

The benchmark in Table 2 shows that the algorithms give very similar modularity values within each dataset, meaning the custom implementations recover communities of comparable quality to the library methods. Library Leiden is usually slightly higher, but the differences are small.

The largest differences appear in runtime. Library Leiden is consistently the fastest method, often by a very large margin. Custom Leiden is the slowest, especially on Pearson graphs, while Custom Louvain is faster than Custom Leiden but still much slower than the optimized libraries.

For NMI and Jaccard, Library Leiden has the strongest agreement with the NetworkX Louvain reference among the non-reference algorithms. Custom Leiden generally agrees more with the reference than Custom Louvain, but the gap is moderate. Since NetworkX Louvain is the reference, its NMI and Jaccard are always 1.000 by definition.

The number of communities is broadly consistent across methods within each dataset, which suggests that all algorithms detect similar partition scales. Small differences remain: Library Leiden often finds slightly more communities, while Custom Louvain sometimes produces fewer.

The within-community and between-community correlation metrics are also very stable across algorithms. This suggests that the detected communities have similar structural meaning, even when the exact node assignments differ slightly.

The main difference between the custom implementations and the library algorithms is the computational efficiency. The custom Louvain implementation is written in pure Python and repeatedly scans the neighbors of a node for each candidate community when computing modularity gain. In contrast, NetworkX first aggregates the edge weights from a node to each neighboring community, so each possible move can be evaluated much faster. This difference becomes especially important for dense Pearson graphs, where each node has many neighbors.

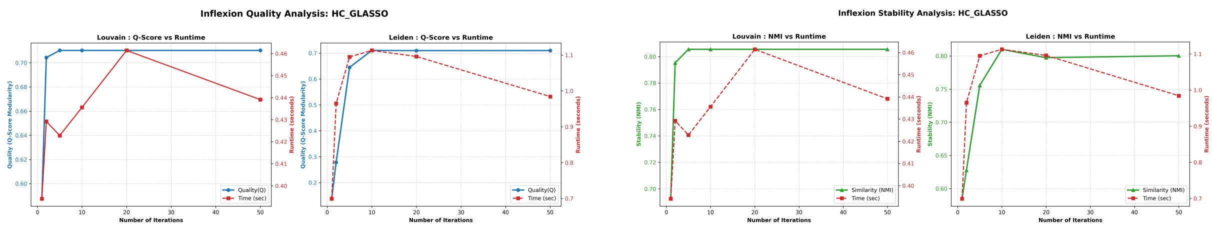
The same issue explains the slower custom Leiden runtime. Although the algorithm reaches comparable modularity and agreement scores, it performs many operations through NetworkX graph objects and Python loops, while Library Leiden relies on optimized igraph/leidenalg routines implemented in compiled code. Therefore, the custom algorithms are useful for transparency and validation, but the library implementations are more appropriate for large-scale benchmarking or repeated experiments.

Dataset	Algorithm	Modularity	NMI	Jaccard	Runtime (s)	Communities	Within Corr	Between Corr	W/B Ratio
hc_glasso	Custom Leiden	0.709	0.821	0.581	0.392	45.2	0.0074	0.0002	45.1
hc_glasso	Custom Louvain	0.710	0.806	0.584	0.161	45.2	0.0073	0.0002	45.5
hc_glasso	Library Leiden	0.721	0.825	0.612	0.030	47.2	0.0078	0.0002	49.2
hc_glasso	NX Louvain	0.713	1.000	1.000	0.045	45.4	0.0075	0.0002	46.5
hc_pearson	Custom Leiden	0.139	0.771	0.738	70.360	4.2	0.3363	0.1795	1.9
hc_pearson	Custom Louvain	0.139	0.712	0.679	34.338	3.6	0.3235	0.1788	1.9
hc_pearson	Library Leiden	0.140	0.825	0.817	0.982	4.0	0.3336	0.1794	1.9
hc_pearson	NX Louvain	0.139	1.000	1.000	1.834	4.2	0.3334	0.1814	1.9
scz_glasso	Custom Leiden	0.750	0.881	0.641	0.514	280.8	0.0070	0.0001	2422.5
scz_glasso	Custom Louvain	0.751	0.878	0.659	0.088	281.8	0.0072	0.0001	2055.1
scz_glasso	Library Leiden	0.757	0.888	0.688	0.022	281.8	0.0072	0.0001	2133.2
scz_glasso	NX Louvain	0.753	1.000	1.000	0.031	281.6	0.0071	0.0001	2127.5
scz_pearson	Custom Leiden	0.120	0.795	0.776	69.456	4.0	0.3139	0.1818	1.8
scz_pearson	Custom Louvain	0.120	0.824	0.790	36.055	3.6	0.3023	0.1831	1.7
scz_pearson	Library Leiden	0.121	0.869	0.858	0.902	4.0	0.3141	0.1826	1.8
scz_pearson	NX Louvain	0.120	1.000	1.000	1.979	4.0	0.3118	0.1830	1.8

Table 2: Algorithm Performance and Accuracy Benchmark across Datasets

5.6 Algorithmic Convergence and the Iteration Ceiling Effect

To optimize execution parameters, we tracked modularity (Q), structural stability (NMI), and runtime across a dynamic iteration ceiling $I_{\max} \in \{1, 2, 5, 10, 20, 50\}$. The empirical convergence profiles are detailed in Figure 9.



(a) Quality Convergence (Q-Score) (b) Structural Stability (NMI)

Figure 9: Parametric benchmarks for HC GLasso networks. Both modularity (blue) and partition stability (green) reach a strict horizontal plateau at $I_{\max} = 10$, rendering higher boundaries redundant relative to runtime (red dashed lines).

The benchmarks reveal an abrupt inflexion point followed by a strict asymptotic plateau. Both Custom Louvain and Custom Leiden reach full mathematical saturation at $I_{\max} = 10$. Beyond this threshold, expanding the iteration limit yields no structural changes, and the partition similarity (NMI) locks at 1.0.

This ceiling effect is driven by the native early stopping mechanisms of the algorithms, where $I_{\max}$ acts as a defensive boundary rather than a fixed cycle count:

- **Louvain:** Exits pre-emptively when the macro-level modularity gain drops below a strict numerical threshold ($\Delta Q < 10^{-7}$).
- **Leiden:** Terminates the refinement phase automatically as soon as no further nodal displacement can minimize the network's structural energy.

On our 1017-node connectomes, both implementations naturally achieve local equilibrium within 3 to 5 internal cycles. Setting $I_{\max} = 50$ does not induce additional computational passes; the algorithms break early once convergence is secured.

The slight negative slope in runtime observed at higher caps is a statistical artifact of the small benchmark sample ($N = 5$) and minor background system fluctuations, as the actual computational workload remains strictly constant once partitions freeze. Consequently, a fixed constraint of $I_{\max} = 10$ is established as the optimal operational trade-off, securing maximum topological stability while minimizing system runtime.

6 Visualisation

The visualisation module produced four families of figures:

- **Distribution plots:** violin + swarm plots of per-subject Q, ARI and NMI by group and method, showing full distributions.
- **3D brain maps:** glass-brain renderings (via `nilearn`) of community structure and bridge nodes, placed at true MNI coordinates computed from atlas parcel centroids.
- **Bridge bar charts:** ranked top-15 bridge ROIs per method, colored by HC reference community membership.
- **Healing curves:** Q and ARI trajectories as a function of the fraction of spurious edges removed, per subject and averaged.

7 Discussion

Our central finding is that schizophrenia preserves the overall amount of network segregation (Q unchanged) while reorganising community membership (ARI/NMI reduced).This directly motivates the bridge analysis: if the amount of structure is preserved but its layout is altered, then the pathology lies in specific mis-routed connections, which the bridge analysis localises. The divergence between GLasso and Pearson bridge rankings demonstrated empirically why partial-correlation estimators are preferred for connectivity analysis: they suppress susceptibility-artefact-driven edges that contaminate raw correlations, and for this, the dual-method design proved to be valuable for the robustness check.

7.1 Limitations

- Effect sizes are moderate; these features are not diagnostic at the single-subject level.
- The healing simulation is a computational thought experiment, not a model of clinical intervention.
- Community detection was performed at a single resolution ($\gamma = 1.0$); a multi-resolution analysis would reveal structure at different scales.
- Age and sex differences between groups were not regressed out of the network metrics.
- The group-average HC reference graph used for visualization (and as a baseline for ARI/NMI computation) is artificially dense compared to individual connectomes due to element-wise averaging, which inherently depresses absolute partition-overlap scores.
- Mathematical Over-Sparsification and Jitter Noise: The use of cross-validation (`GraphicalLassoCV`) introduces a systematic density contrast compared to Pearson correlation. Due to the high multi-collinear and noisy nature of fMRI time series, the cross-validation criteria aggressively prioritizes a large penalty parameter λ to stabilize matrix inversion across folds. This numerical conservative mechanism suppresses weaker or indirect pathways entirely, potentially over-filtering long-range polysynaptic connections that might hold biological significance.

7.2 Future Work

Several extensions would strengthen and broaden the present analysis.

- Larger-scale healing simulation. For computational tractability the simulated-healing experiment was run on a representative subsample of ten patients. Running it on all 72 SCZ subjects would tighten the confidence around the mean recovery curve and allow the healing trajectory itself to be tested statistically (for example, whether the edge-removal fraction at which SCZ ARI crosses the healthy baseline differs systematically across patients).

- Independent replication. The findings are derived from the single COBRE cohort. Replicating the full pipeline on an independent dataset such as fBIRN or the Human Connectome Project Early-Psychosis sample would establish whether the bridge-node localisation (Frontoparietal Control, Default-Mode, Dorsal Attention) generalises beyond this cohort, which is the standard bar for a neuroimaging biomarker claim.

- Multi-resolution community detection. All community detection was performed at the default resolution $\gamma = 1.0$. Because brain networks are hierarchical, sweeping γ across a range would reveal community structure at multiple scales and test whether the SCZ reorganisation is specific to one scale or pervasive across them.

- Statistical validation of community structure. A degree-preserving permutation test (random rewiring via double-edge swaps) would confirm that the observed modularity exceeds the null distribution of randomly rewired graphs, ruling out the possibility that the detected communities are artefacts of the algorithm finding spurious structure in noise.

- Covariate adjustment. Age, sex and head-motion differences between groups were not regressed out of the network metrics. Incorporating these as covariates would isolate the disease-specific contribution to the observed ARI/NMI differences from demographic and motion confounds.
- Linking to clinical phenotype. Finally, relating each patient's ARI deviation to clinical scores (symptom severity, illness duration, medication load) would test whether the network reorganisation tracks the clinical expression of the disorder, moving the analysis from group-level characterisation toward individual-level relevance.

8 Conclusion

We built an end-to-end network-analytics pipeline that ingests preprocessed fMRI connectivity matrices, detects communities with custom Louvain and Leiden implementations, and compares patients to a healthy reference. We found that schizophrenia reorganises brain network communities without changing their overall modularity, localised the responsible connections to higher-order cognitive networks via a bridge-edge analysis, and demonstrated their causal role through a simulated healing experiment. Running the entire pipeline under two connectivity estimators and two community-detection algorithms established both the robustness of the scientific findings and the methodological importance of the connectivity-estimation choice.

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A Louvain Replication

All main-text analyses were repeated using the Louvain algorithm in place of Leiden. The conclusions are identical: modularity does not distinguish groups, ARI and NMI are significantly lower in SCZ, the GLasso bridge nodes are higher-order cognitive regions, and the simulated healing restores ARI toward the healthy baseline. This confirms that the findings are not artefacts of the particular community-detection algorithm. Figures 10 through 15 reproduce the main-text figures using Louvain.

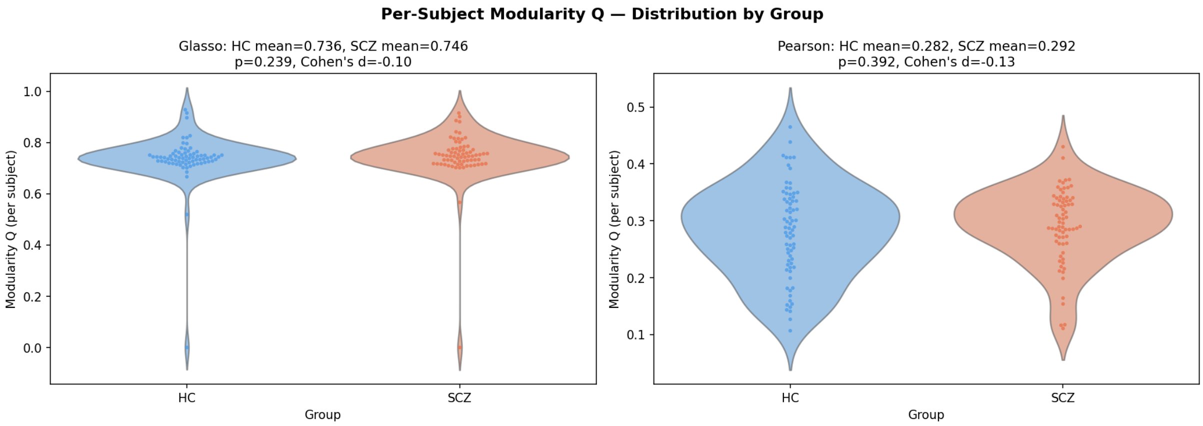

Figure 10: Per-subject Q by group (Louvain). Cf. Figure 3.

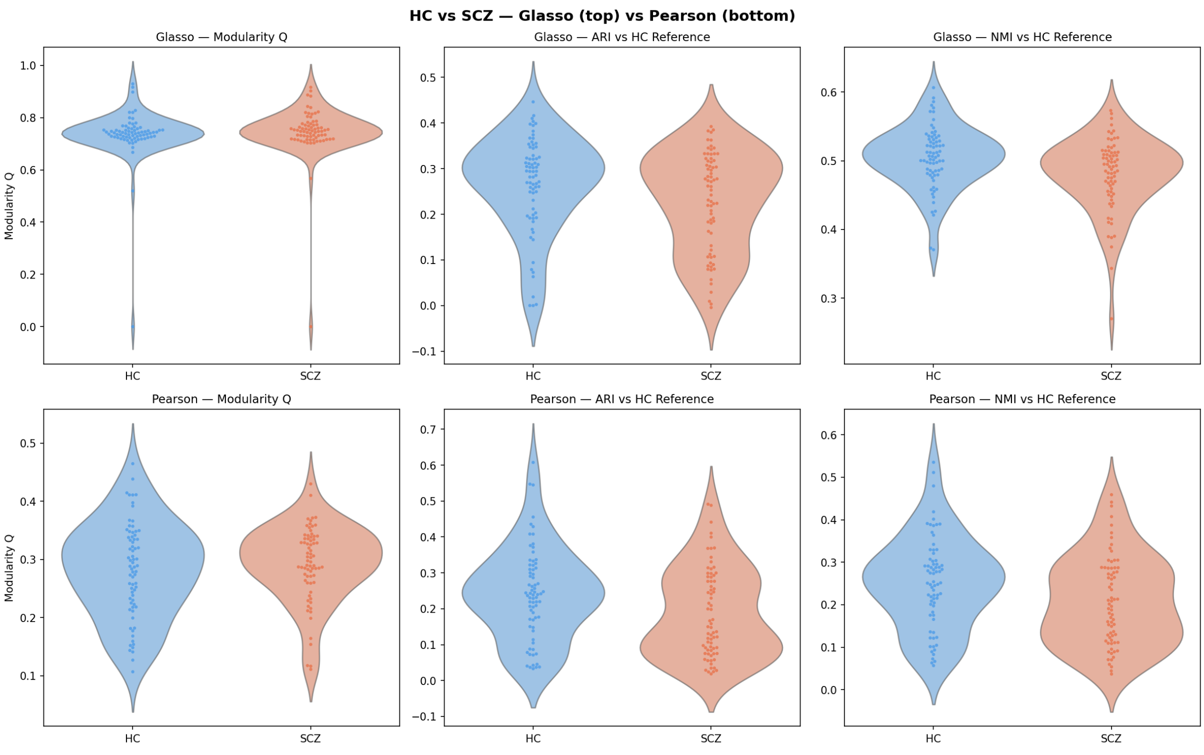

Figure 11: Q/ARI/NMI distributions (Louvain). Cf. Figure 4.

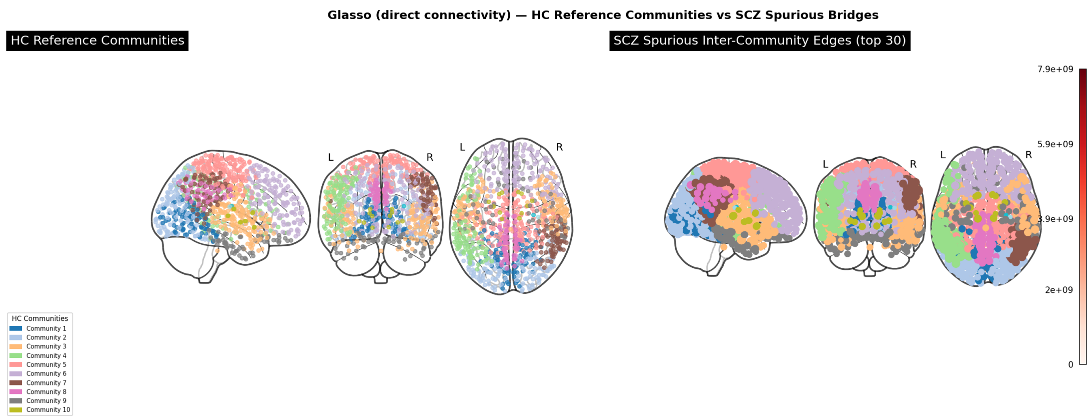

Figure 12: GLasso brain map (Louvain). Cf. Figure 5.

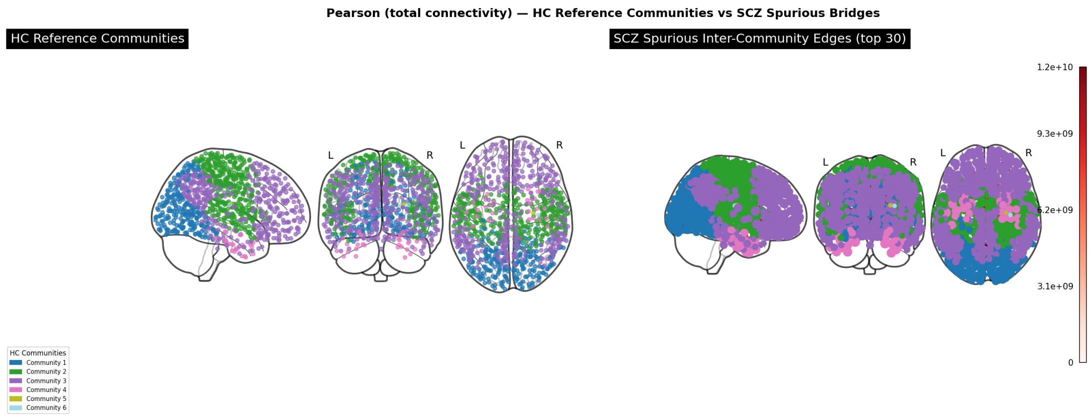

Figure 13: Pearson brain map (Louvain). Cf. Figure 6.

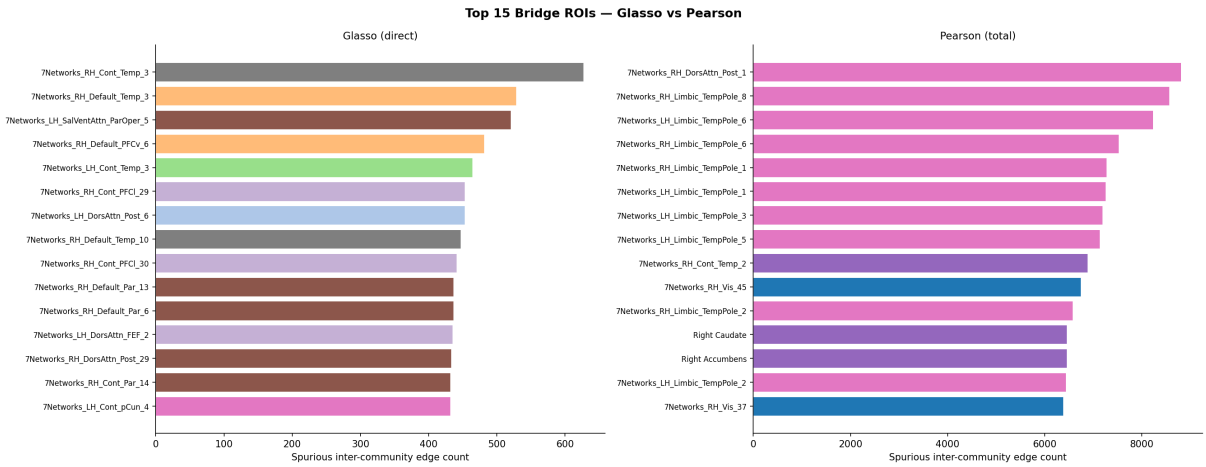

Figure 14: Top-15 bridge ROIs (Louvain). Cf. Figure 7.

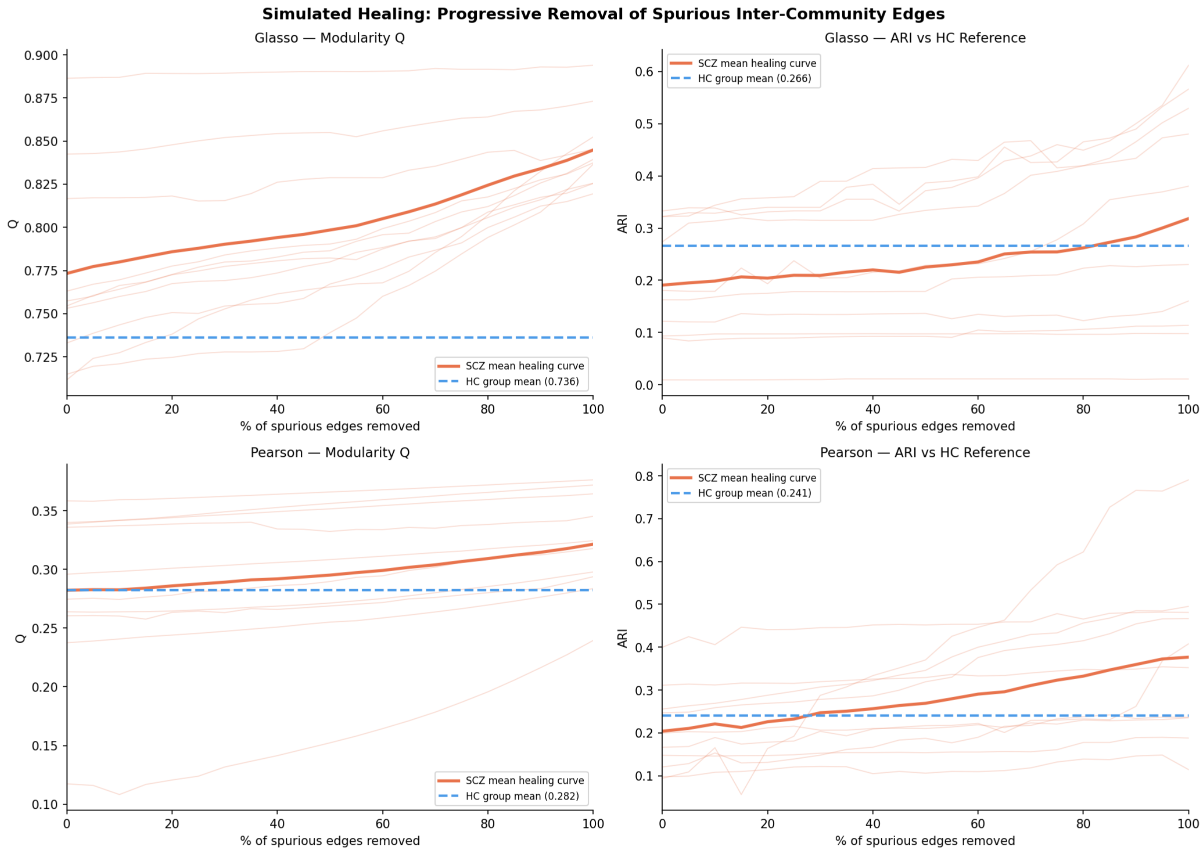

Figure 15: Simulated healing (Louvain). Cf. Figure 8.